



Contents lists available at ScienceDirect

International Immunopharmacology

journal homepage: www.elsevier.com/locate/intimp



Herbal Formula-3 inhibits food allergy in rats by stabilizing mast cells through modulating calcium mobilization



Bo Yang^{a,d,1}, Jianjie Li^{a,d,1}, Xiaoyu Liu^{a,d}, Li Ma^{a,d}, Li Deng^b, Jie Liu^c, Zhigang Liu^{a,d}, Qiongmei Ji^{d,*}

^a State Key Laboratory of Respiratory Disease for Allergy, School of Medicine, Shen Zhen University, Shen Zhen 518060, China

^b College of Life Sciences, Shen Zhen University, Shen Zhen 518060, China

^c Department of Pathophysiology, School of Medicine, Shen Zhen University, Shen Zhen 518060, China

^d Institute of Allergy and Immunology, School of Medicine, Shen Zhen University, Shen Zhen 518060, China

ARTICLE INFO

Article history:

Received 4 March 2013

Received in revised form 13 May 2013

Accepted 9 June 2013

Available online 5 July 2013

Keywords:

Chinese herbal medicine

Food allergy

Mast cell

Degranulation

Calcium

Store-operated calcium channels

ABSTRACT

Background: The prevalence of food allergy has increased dramatically during the last three decades, but currently there was no effective therapy except avoidance of allergen. This study aimed to investigate the effect of a modified Chinese herbal formula, Formula-3, on mast cell degranulation and the anti-allergic activity in both animal and cell models.

Methods: With OVA-sensitized food allergic model in Brown–Norway rats, we checked tissue injury in the small intestines by H&E staining. The Th2 cytokine levels and IgE production in serum or supernatant of the intestinal mucosa homogenates were analyzed by ELISA. Meanwhile, rat peritoneal mast cell activation and degranulation were examined by Toluidine Blue Stain and the release of histamine was measured. Furthermore, the regulation of Formula-3 on Ca^{2+} mobilization was investigated by probing intracellular Ca^{2+} with fluo-4 fluorescence. The direct effect of Formula-3 on mast cell stabilization was also studied in RBL-2H3 cell line.

Results: In vivo Formula-3 administration significantly reduced tissue damage in the small intestines of rat and suppressed Th2 cytokine secretion and IgE production. We demonstrated that Formula-3 treatment significantly suppressed FcεR1-mediated mast cell degranulation no matter in OVA-challenged allergic rats or IgE-sensitized RBL-2H3 cell line. Furthermore, Formula-3 significantly decreased Ca^{2+} influx through store-operated calcium channels (SOCs) evoked by dinitrophenyl-BSA or thapsigargin in mast cells.

Conclusion: Taken together, our data indicate that Formula-3 stabilizes mast cells by suppressing FcεR1-induced Ca^{2+} mobilization mainly through inhibiting Ca^{2+} entry via SOCs, thus exerting a protective effect against OVA-sensitized food allergy.

© 2013 Published by Elsevier B.V.

1. Introduction

Food allergy, mediated by IgE or non-IgE reaction [29], is an adverse health effect arising from a specific immune response that occurs reproducibly on exposure to a given food [5]. During the last several decades, the prevalence of food allergies has increased dramatically worldwide [14,27,31]. The prevalence of food allergy to milk, eggs and peanuts is reported to be around 6–8% of children under the age of three [13,34], while it is less common in adult population with a percentage of about 4% [43]. Although any food can trigger an allergic response and more than 170 foods have been reported to cause IgE-mediated reaction, 90% of food allergy is caused by the “eight big food allergens”, including peanut, tree nuts, egg,

milk, fish, crustacean shellfish, wheat, and soy [4]. IgE-mediated reactions are characterized by an acute onset of symptoms generally within 2 h after ingestion of or exposure to the trigger food [5].

At the present time, unfortunately, there were no effective therapies for IgE-mediated food allergies and avoidance of allergen may be the only way for these patients. Taken in consideration of the side effect from “lifelong” nature of allergy and chronic use of current drugs, around 30% of adults and 60% of children seek to use complementary and alternative medicine (CAM) treatment for a variety of diseases, including asthma [36]. For example, it has been reported that *Ganoderma tsugae* supplementation could attenuate airway inflammation and Th2 response in allergic asthma in the murine model [7]. Compound K, one main metabolite from Panax ginseng, has been demonstrated to have potent inhibitory activity on β-hexosaminidase release from RBL-2H3 cells and on the passive cutaneous anaphylaxis (PCA) reaction [9]. *Angelica acutiloba* inhibited mast cell-derived allergic reactions by inhibiting the release of histamine and the production of pro-inflammatory cytokines, which is dependent on the phosphorylation of ERK and JNK [22]. Ginger can

* Corresponding author at: Rm722, Composite Building, School of Medicine, Shen Zhen University, Nan Hai Ave 3688, Shen Zhen, China. Tel.: +86 755 86671907; fax: +86 755 86671906.

E-mail address: jqm196910@126.com (Q. Ji).

¹ Both authors have contributed equally to this work.

suppress Th2-mediated immune responses and serum IgE levels in a mouse model with airway inflammation [1]. Li et al. have recently reported that Chinese herbal Formula-2 [25] and -1 [26] could reverse IgE-mediated allergic hypersensitivity associated with reduction of Th2 responses in a murine model of allergic asthma or peanut allergy respectively. However, the underlying mechanism concerning Chinese herbal medicine therapy on food allergy remains unknown.

In the present study, we for the first time investigated the therapeutic effect of a modified Chinese herbal medicine, Formula-3, on the treatment of OVA-mediated food allergy in rats. Furthermore, we found that the underlying mechanism was related to mast cell degranulation and Ca^{2+} mobilization through store-operated calcium channels (SOCs).

2. Materials and methods

2.1. Formula-3 preparation

Modified Chinese herbal medicine, Formula-3, consists of 6 herbs (Table 1). The daily human adult dosage (96.0 g; Table 1) of this formula was decocted (water-extracted) as previously reported [26]. Briefly, all the components were boiled for 2 h, and the decoction was filtered and lyophilized. Through use of a conversion table of equivalent effective dose ratios from human beings to animals based on body surface area [26], 150 mg per rat of lyophilized decoction in 1 ml was administered to each rat daily.

2.2. Animals

Four week old female Brown–Norway (BN) rats were purchased from Vital River Laboratories (Beijing, China) and housed in groups of four per cage in a controlled environment with a photoperiod of 12 h light–12 h dark and a temperature of $20 \pm 2^\circ\text{C}$. Sanitary controls were performed for all major rodent pathogens, and the results of these tests were uniformly negative. All the animal experimental procedures were approved by the Animal Care and Use Committee of Shenzhen University and carried out in accordance with the Guide for the Care and Use of Laboratory Animals published by the US National Institutes of Health (NIH publication no. 85-23, revised 1996).

Forty-eight BN rats were randomly divided into four groups: control group, ovalbumin (OVA) group, OVA + Formula-3 group and Formula-3 group. Each group received PBS, OVA, or Formula-3 as shown in Fig. 1 [21]. Control group received 1 ml PBS (0.1 M) daily by gavage administration for 13 weeks, while OVA group was orally treated with 1 ml OVA (1 mg/ml) for the first 6 weeks and the 7th, 9th, 11th, and 13th weeks, and the mice were given with 1 ml PBS (0.1 M) in 8th, 10th, and 12th weeks. OVA + Formula-3 group received Formula-3 (150 mg/ml $\times$ 1 ml daily per rat) oral treatment daily from Week 7 to 13. Formula-3 group was not sensitized by OVA. All groups of rats were stimulated by 100 mg/ml OVA for 24 h at the end of 13th week and then sacrificed for the following experiments.

Table 1

The recipes of modified Chinese herbal medicine, Formula-3.

	TCM materia medica (pinyin)	Equivalent pharmaceutical name	Amount (g)	Part used
1	Ling zhi	Ganoderma lucidum	30	Fruiting body
2	Wu mei	Fructus pruni mune	30	Fruit
3	Huang lian	Rhizoma coptidis	9	Root
4	Gan jiang	Rhizoma zingiberis officinalis	9	Root
5	Ren shen	Radix ginseng	9	Root
6	Dang gui	Corpus radix angelicae sinensis	9	Root
Total			96	

References: Li [25,26].

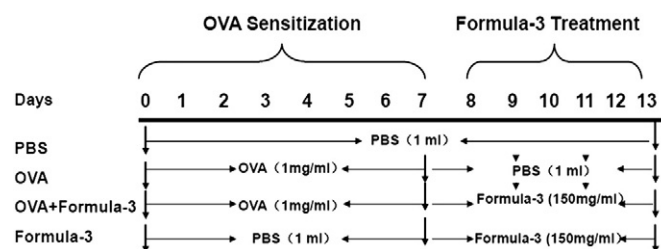


Fig. 1. Protocol of OVA sensitization/challenge and PD treatment. Rats were sensitized with OVA (1 mg/ml $\times$ 1 ml daily per rat) intragastrically for 48 days. For PD treatment group, the rats were orally treated with PD (30 mg/ml $\times$ 1 ml daily) from Day 49 to 52 after OVA sensitization.

2.3. H&E staining in the small intestine tissue

The intestine jejunum part of the small intestine was isolated from rats and embedded with paraffin. 7 μm sections were prepared and subjected to H&E staining as previously reported [8].

2.4. Toluidine Blue Stain

Typical mast cells in rat small intestine tissue or peritoneal lavage solution (RPLS) were stained with Toluidine Blue Stain as previously described [12]. Briefly, 200 μl peritoneal lavage fluids were dried in the air on cromolyn sodium pretreated slides and then covered with several drops of staining solution (Toluidine Blue Stain dissolved in 70% ethanol). After 90 s, the staining solutions were washed away quickly with running tap water and the stained cells were examined and counted under light microscope (Olympus, Japan).

2.5. Enzyme-linked immunosorbent assay (ELISA)

The contents of interleukin-4 (IL-4), interleukin-10 (IL-10), and histamine in rat RPLS and serum were assayed by commercial ELISA kits using paired antibodies according to the manufacturer's instructions. The kits for detecting IL-4 and IL-10 were bought from eBioscience (USA) and the kit for histamine was bought from R&D Inc. (USA). Serum IgE levels were also checked using a commercial ELISA kit (BD Pharmingen, USA) following the manufacturer's instructions.

2.6. Rat peritoneal mast cell isolation

The BN rats were sacrificed after anesthetizing by inhalation of ether in air. Rat peritoneal mast cells (RPMCs) were obtained by peritoneal lavage and purified by density gradient fractionation as described previously [42,44]. Isolated RPMC preparations contained >98% mast cells and at least 98% of these cells were viable as checked by metachromatic staining in 0.05% Toluidine Blue.

2.7. Ca^{2+} imaging by confocal microscope

Intracellular Ca^{2+} signal was measured as described previously with minor modification [10]. RPMCs or RBL-2H3 cells were incubated with 5 $\mu\text{mol/l}$ Ca^{2+} fluorescent probe fluo-4 AM (Invitrogen, CA, USA) for 30 min at room temperature. After washing with Tyrode's buffer three times, the dye inside the cells was allowed to deesterify for 30 min at 37°C . It has been determined that nearly 95% of the fluorescent dye was retained in the cytoplasm. Fluorescent images of Ca^{2+} were obtained using an Olympus 1000 confocal microscope with $40\times$ oil immersion lens (NA 1.3) (Olympus, Japan). Fluo-4 signal was excited at 488 nm and emitted at >505 nm. Frame-scan images were acquired at sampling rate of 15 ms per frame and 20 s per interval.

Image data were analyzed off-line using the FV10-ASW 2.1 software. A selected image from each image set was used as a template for designating the region of interest (ROI) within each cell. The

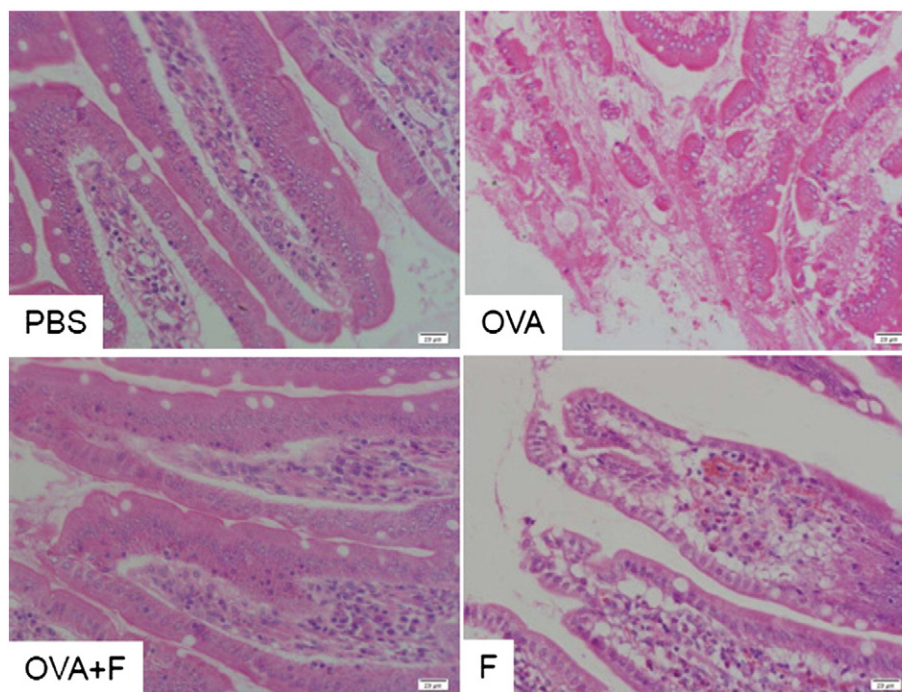


Fig. 2. Formula-3 attenuated tissue injury in the small intestine caused by OVA sensitization. Morphology of intestine jejunum was analyzed by H&E staining. Representative images from three independent experiments are shown. Magnification, 400 $\times$. OVA, ovalbumin; F, Formula-3.

integrated intracellular Ca^{2+} concentration was determined by calculating $\Delta F/F_0$. F_0 was defined as the mean basal fluorescence intensity of the dye recorded during the first 5–10 scanning frames, when the cells were under rest conditions. ΔF denotes ($F-F_0$), where F is the temporal fluorescence intensity. The $\Delta F/F_0$ values within each ROI were plotted as a function of time (see Fig. 5 for typical time-courses of Ca^{2+} response to thapsigargin (TG) or dinitrophenyl (DNP)-BSA stimulation in single RBL-2H3 cell). The amplitude of the Ca^{2+} response within each cell was quantified as the highest $\Delta F/F_0$ level reached during the measurement period, which was averaged over all cells within each group.

2.8. Cell viability assays

RBL-2H3 cell viability was assayed by 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT, Roche Applied Science) as described previously [45]. The cells were incubated with 0.5 mg/ml MTT for 4 h, and the absorbances at 490 nm were measured.

2.9. Degranulation assay

Degranulation was monitored by the release of β -hexosaminidase, as previously described [17]. Briefly, cultured RBL-2H3 cells in 96-well plates (2×10^4 cells/well) were sensitized overnight with 0.5 $\mu\text{g}/\text{ml}$ DNP-specific IgE (Sigma Aldrich, USA). The cells were subsequently washed twice with PBS and stimulated with 250 ng/ml of DNP-BSA (Sigma) in Tyrode's buffer at 37 $^\circ\text{C}$ for 30 min, and mixed with an equal volume of substrate solution (*p*-nitrophenyl-N-acetyl- β -D-glucosaminide, 2 mM in 0.1 M citrate, pH 4.5) and incubated for 1 h at 37 $^\circ\text{C}$. The reaction was then terminated by adding 200 μl of stopping buffer (0.1 M $\text{Na}_2\text{CO}_3/\text{NaHCO}_3$, pH 10.0) and the absorbance was measured with a microplate reader (BioRad Company, USA). Cell was lysed with 0.1% Triton X-100, and the β -hexosaminidase activity of the extracts was measured (total). Percentage degranulation to each stimulation was calculated as a portion of β -hexosaminidase release in the culture medium to the whole β -hexosaminidase release in the culture medium and in the cell lysate.

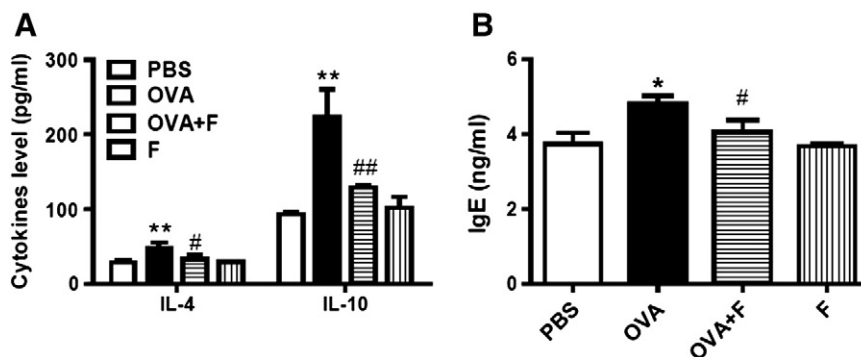


Fig. 3. Formula-3 suppressed Th2 cytokine levels and IgE production in OVA-allergic rats. A, The cytokine levels in RPLS were analyzed by ELISA. B, Statistical analysis of OVA-specific IgE in serum, which was collected from allergic rats administered with or without Formula-3. $N = 8$. *, $P < 0.05$; **, $P < 0.01$ vs PBS group. #, $P < 0.05$; ##, $P < 0.01$ vs OVA group. OVA, ovalbumin; F, Formula-3.

2.10. Statistics

Data are presented as means $\pm$ SEM. When two comparisons were obtained, Student's unpaired two tailed *t* test was used. When multiple comparisons were obtained, the analyses consisted of one-way ANOVA for repeated measures and Student–Newman–Keuls multiple comparison test. A value of $P < 0.05$ was considered to be statistically significant.

3. Results

3.1. Formula-3 attenuated OVA-challenge-caused tissue damage in the small intestine of rats

In the present study, we used 1 mg OVA oral sensitization to establish food allergic model in BN rats as previously reported [18,21]. The tissue damage in the intestine was checked by H&E staining as

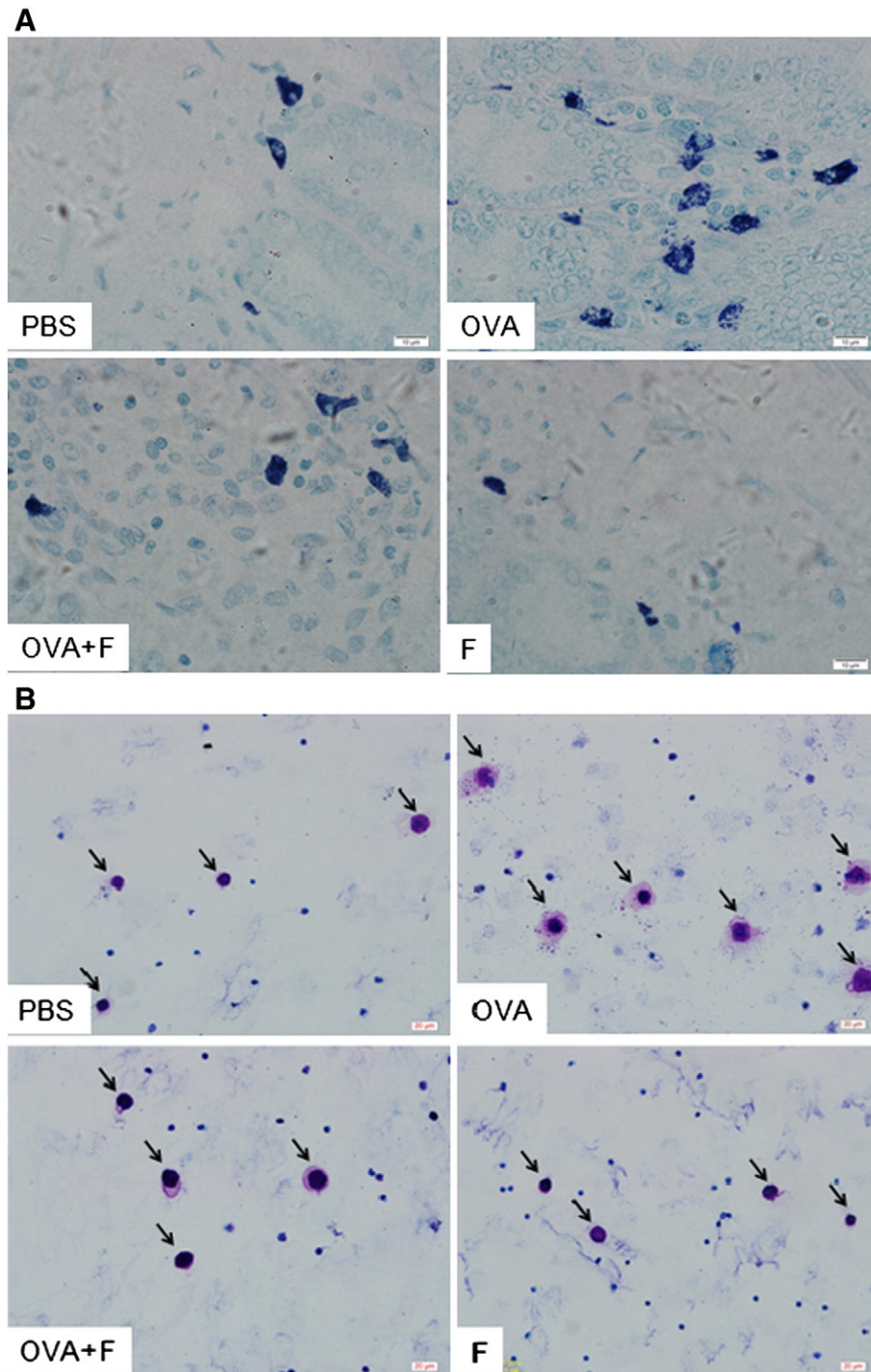


Fig. 4. Formula-3 therapy reduced mast cell degranulation and activation. The mast cells in intestinal mucosa (A) or RPLS (B) were identified by Toluidine Blue Stain. Magnification, 400 $\times$. Mast cells were considered degranulated if at least 5 granules appeared outside the cell body. Arrows indicate mast cell. C, Degranulated mast cells in intestinal mucosa (left panel) or RPLS (right panel) were counted from at least 200 total cells and the percentage of degranulation was calculated as degranulated cells against total cells. D, The release of histamine in serum (left panel) or RPLS (right panel) was measured by ELISA. $n = 8$. *, $P < 0.05$; **, $P < 0.01$ vs PBS group. #, $P < 0.05$; ##, $P < 0.01$ vs OVA group.

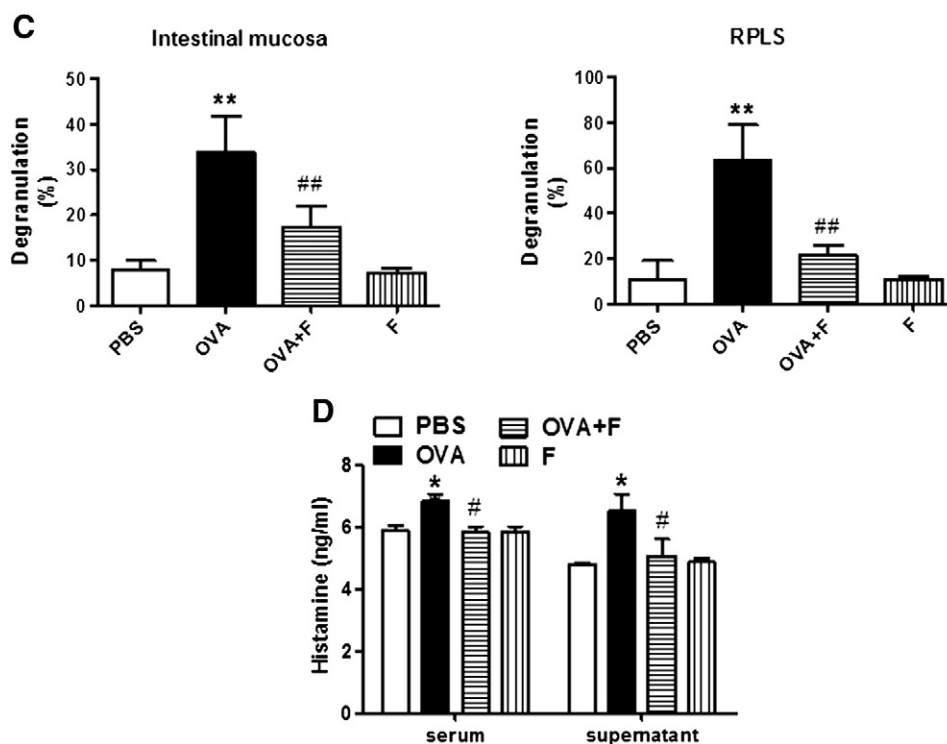


Fig. 4 (continued).

shown in Fig. 2. The results showed that the intestinal mucosa was severely injured, the organization of intestinal villus was in a mess and the numbers of intestinal villus were significantly reduced in OVA group, which was attenuated by Formula-3 treatment. Administration of Formula-3 also reduced inflammatory cell infiltration into the mucosa and the increase of submucosa thickness.

3.2. Treatment with Formula-3 decreased Th2 cytokine levels and attenuated IgE and histamine release in OVA-sensitized group

The cytokine levels in the supernatant of the intestinal mucosa were measured by ELISA. The results showed that type Th2 cytokines, including IL-4 and IL-10 in OVA-challenged group were significantly higher than those in control group (IL-4, 48.79 ± 6.65 vs Ctrl 28.67 ± 2.54 pg/ml; IL-10, 223.83 ± 36.57 vs 93.00 ± 3.54 pg/ml) (Fig. 3A). Treatment with Formula-3 reduced the enhancement of IL-4 and IL-10 by 32% and 43% respectively. Meanwhile, ELISA analysis showed that the concentration of OVA-specific IgE in serum was enhanced by 1.3 fold in OVA group and Formula-3 therapy reversed it to normal level (Fig. 3B).

3.3. Formula-3 reduced mast cell activation and degranulation in the small intestine

Mast cell degranulation and histamine release are major events in food allergy. The number and morphology of the mast cells in rat small intestine tissues (Fig. 4A) or RPLS (Fig. 4B) were examined by Toluidine Blue Stain. In OVA group, the number of mast cells was significantly increased and the cell size was much bigger, with more shrink on the cell membrane, bubbles in the cytoplasm, and degranulation of vesicles around the cells. In vivo administration with Formula-3 for 6 weeks reversed OVA-challenge-induced damage in mast cells. The ratio of mast cell degranulation as indicated by vesicles (at least five) around the cells was dramatically increased in OVA group by 5.7 fold and fell down by 65% after Formula-3 treatment (Fig. 4B). Formula-3 mediated attenuation of mast cell degranulation was further confirmed

by decreased histamine levels (Fig. 4D). It has been found that histamine release in both serum and RPLS was significantly increased in OVA-induced food allergic group, which was attenuated by Formula-3 therapy by ~15% and 22% respectively.

3.4. Formula-3 reduced Ca^{2+} entry through SOCs in food allergic mast cells

It has been suggested that an increase in intracellular Ca^{2+} through SOCs is essential for mast cell degranulation [11]. We have previously found that food allergen-induced mast cell activation is related to stimulation of Ca^{2+} mobilization as well (data not published). To explore the direct effects of Formula-3 on Ca^{2+} entry through SOCs, we used a standard Ca^{2+} add-back assay, in which intracellular Ca^{2+} stores were depleted by TG, a sarcoplasmic/endoplasmic reticulum calcium ATPase (SERCA) (Ca^{2+} pump) blocker, in Ca^{2+} -free extracellular solution, after which the extracellular Ca^{2+} concentration was returned to 2 mM [2]. Using this protocol, TG elicited two Ca^{2+} peaks, where the first one represented the ER Ca^{2+} release and the second represented Ca^{2+} entry through activated SOCs. As shown in the first peaks in Fig. 5A–D, when the cells were in Ca^{2+} -free solution, the TG-evoked Ca^{2+} amplitude was similar in all the groups, suggesting that the amounts of Ca^{2+} released from ER are nearly the same. In the presence of 2 mM extracellular Ca^{2+} , the TG-evoked Ca^{2+} influx was dramatically enhanced in OVA-sensitized RPMC by 1.5 fold, while Formula-3 treatment reduced the Ca^{2+} entry to normal level. The results indicate that Formula-3 attenuated OVA-induced Ca^{2+} influx elevation through SOCs.

3.5. Formula-3 inhibited DNP-IgE sensitized RBL-2H3 cell degranulation

In order to test whether mast cells are the main and direct effectors contributing to Formula-3 mediated attenuation of food allergy in BN rats, cultured mast cells are introduced in the following study. RBL-2H3 is a basophilic leukemia cell line, which has been extensively used to study mast cell activation and degranulation, FcεR1 structure

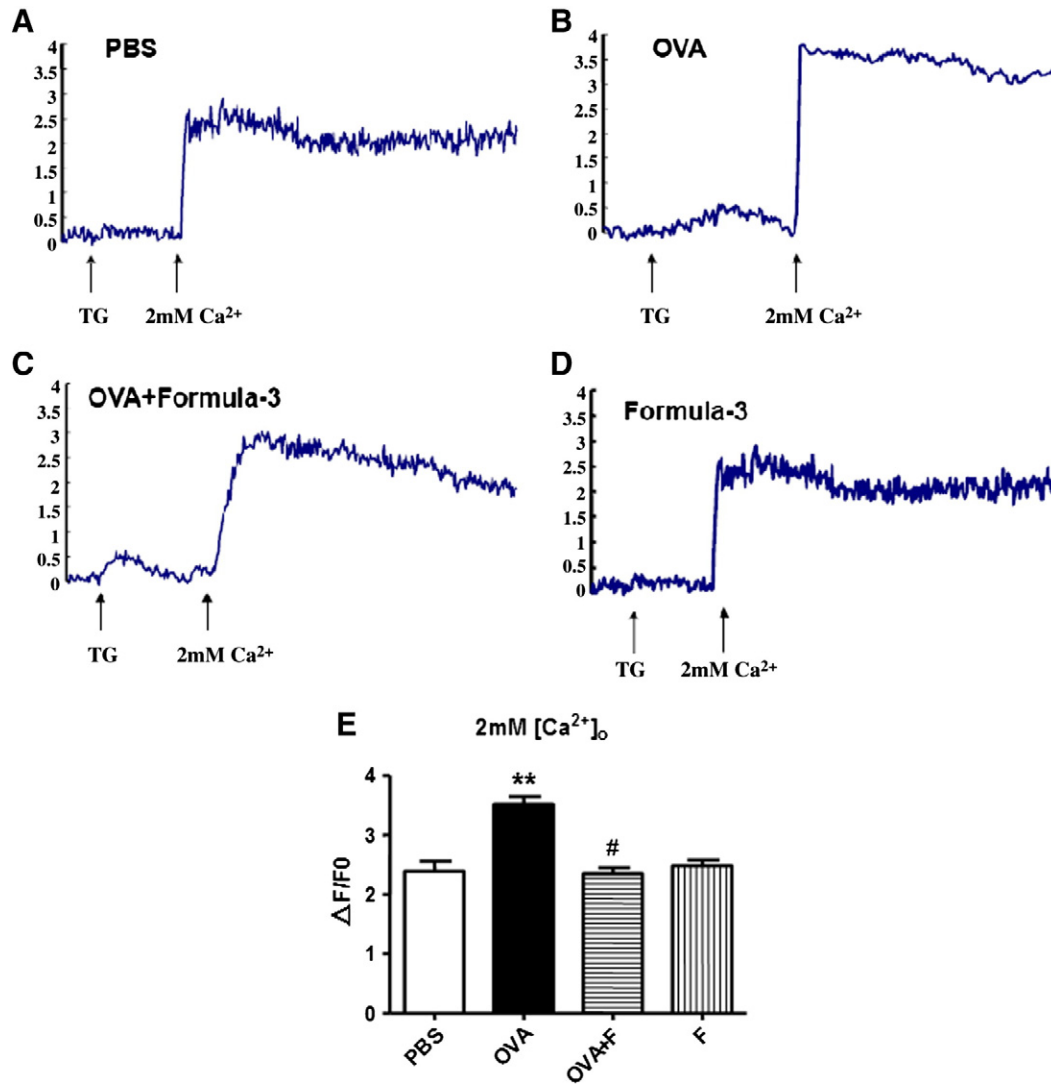


Fig. 5. Formula-3 reduced Ca^{2+} entry through SOCs in food allergic mast cells. RBL-2H3 cells were isolated from BN rat treated with or without Formula-3. Intracellular Ca^{2+} was indicated by fluo-4 fluorescence. Total cell numbers are 40–50 for each group, and the cells were from four independent experiments. Typical responses of TG-evoked Ca^{2+} entry through SOCs in RBL-2H3 cells are shown in A (PBS group), B (OVA group), C (OVA + Formula-3 group), and D (Formula-3 group). E, Averaged peak amplitude of Ca^{2+} entry (second peak) as recorded in each group. **, $P < 0.01$ vs PBS group. #, $P < 0.05$ vs OVA group. F, Formula-3.

and the biochemical pathways [16]. β -hexosaminidase release is a liable marker for mast cell degranulation. It was demonstrated that Formula-3 did not have effect on β -hexosaminidase release in RBL-2H3 cell under basal condition (data not shown). For the following investigation, the RBL-2H3 cells were sensitized with DNP-specific IgE overnight in 1.0 ml of DMEM, and varying doses of Formula-3 [38] were added to the culture medium for 0.5 h or 12 h and challenged with DNP-BSA to induce mast cell degranulation before supernatants were collected. As shown in Fig. 6A, at 0.5 h, treatment with 0.05 mg/ml, 0.2 mg/ml or 0.8 mg/ml Formula-3 significantly reduced β -hexosaminidase release by 15%, 30% and 35% in RBL-2H3 cells, while it attenuated by 25%, 50%, and 60% at 12 h. In consistent with the results in OVA-induced food allergic model, Formula-3 therapy suppressed Th2 cytokine levels in IgE-sensitized RBL-2H3 cells. The cytokine levels in the culture medium of RBL-2H3 cells were measured by ELISA. The results showed that type Th2 cytokines, including IL-4 and IL-10, in positive group (RBL-2H3 cells sensitized with DNP-IgE and challenged with DNP-BSA) were significantly higher than those in control group (IL-4, 18.66 ± 0.19 vs Ctrl 6.82 ± 0.77 pg/ml; IL-10, 63.28 ± 16.69 vs 20.71 ± 1.99 pg/ml). Pretreatment with Formula-3 for 0.5 h significantly decreased Th2 cytokines by 40% in IL-4 and 50% in IL-10 (Fig. 6B). To determine whether Formula-3 suppression of mast cell

degranulation was due to cytotoxicity, we investigated the effects of Formula-3 on cell viability using an MTT assay. The results revealed that treatment with Formula-3 in three dosages (0.05 mg/ml, 0.2 mg/ml, 0.8 mg/ml) for 0.5 h or 12 h had no significant effect on cell viability (Fig. 6C), indicating that Formula-3 suppression of mast cell degranulation could not be attributed to toxicity.

3.6. Formula-3 attenuated Ca^{2+} entry through SOCs in activated RBL-2H3 cells

Mast cell activation is dependent on an increase in cytosolic Ca^{2+} concentration, which is associated with release from intracellular stores and influx of external Ca^{2+} through SOCs. The effect of Formula-3 on intracellular calcium mobilization in RBL-2H3 cells was studied by monitoring intracellular Ca^{2+} fluorescence intensity. In the presence of 2 mM extracellular Ca^{2+} , when the Ca^{2+} store was actively depleted by DNP-BSA, DNP-IgE sensitization induced intracellular Ca^{2+} elevation ($\Delta F/F_0$) was significantly reduced by 35% by treatment with 0.2 mg/ml Formula-3 in RBL-2H3 cells (Fig. 7A and C). Meanwhile, the time to peak of intracellular Ca^{2+} concentration was remarkably extended in Formula-3 treated group (28 ± 5.01 vs Ctrl 12.5 ± 2.05 s) (Fig. 7B and D).

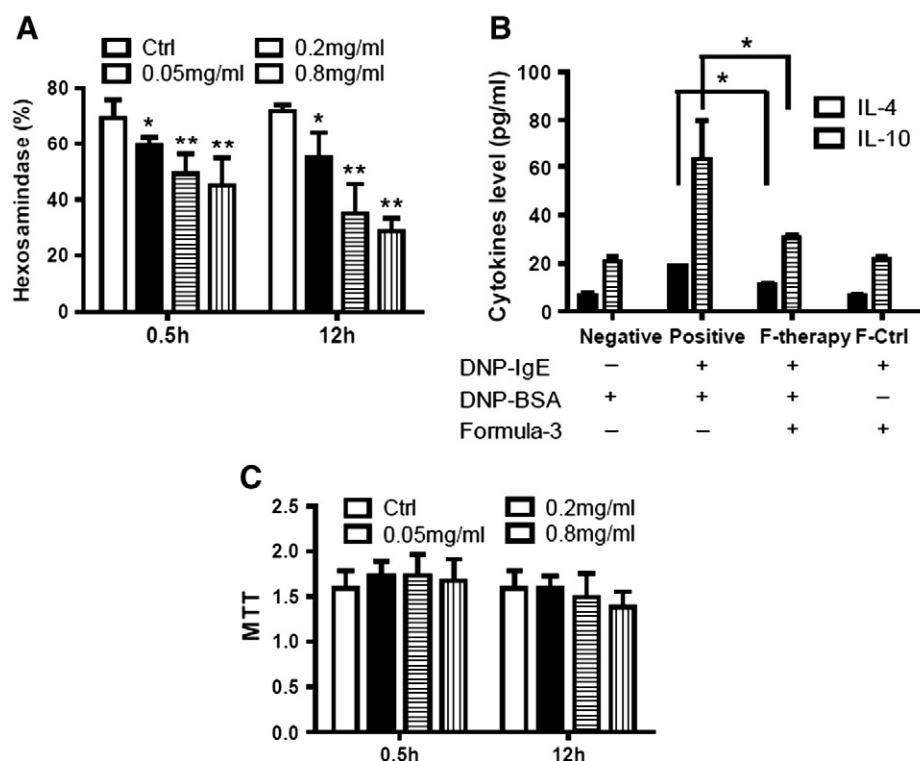


Fig. 6. Formula-3 inhibited β -hexosaminidase release and reduced Th2 cytokine secretion from sensitized RBL-2H3 cells. **A**, The effects of different doses of Formula-3 on mast cell degranulation. Mast cell degranulation was assessed by measuring the relative release of β -hexosaminidase in IgE sensitized RBL-2H3 cells. **B**, The suppressive effect of Formula-3 on Th2 cytokine secretion from IgE-sensitized RBL-2H3 cells. **C**, The effects of different doses of Formula-3 on cell viability by the MTT assay. $N = 8$. *, $P < 0.05$; **, $P < 0.01$ vs Ctrl or positive group. F, Formula-3.

The direct effect of Formula-3 on Ca^{2+} entry in RBL-2H3 cells was investigated by using a standard Ca^{2+} add-back assay as described in Section 3.4. Formula-3 pretreatment significantly decreased the amplitude of Ca^{2+} entry by 36% (Fig. 7G and H), from 1.81 ± 0.16 ($n = 45$) in the control to 1.15 ± 0.37 ($n = 52$) in Formula-3 treated group. This result indicated that Formula-3 had a direct suppressive effect on SOC activity.

4. Discussion

There are two major findings in the present study. First, we identified Formula-3 as a novel mast cell stabilizer, exerting therapeutic effects against food allergy by decreasing antigen-stimulated mast cell degranulation. Second, we demonstrated that Formula-3 suppressed Fc ϵ R1-mediated Ca^{2+} mobilization by inhibiting Ca^{2+} entry through SOCs, which were the major contributors to Formula-3-induced mast cell stabilization.

Food allergy is an immunological adverse reaction caused by food, which encompasses a range of disorders including IgE-mediated anaphylaxis, food protein-induced enterocolitis syndrome, and food-induced eosinophilic gastrointestinal disorders. Allergen from egg is one of the most frequent causes of food allergic reaction as reported [32]. In this study, food allergic model was established in rats by OVA oral sensitization. The allergic animal exhibited enhanced Th2 cytokine levels, OVA-specific IgE concentration and mast cell activation and degranulation.

Emerging data suggest that Chinese herb medicines have potent effects on the treatment of allergic diseases, including asthma [6,15,24], food allergy [33,38–40], allergic rhinitis, and atopic dermatitis. In the present study, Formula-3 which consists of six Chinese herbal medicines reversed OVA-sensitized allergic conditions in both BN rats and RBL-2H3 cell lines. Following administration with Formula-3, the rats did not have apparent anaphylactic symptoms (data not shown) and had a marked decrease in Th2 responsiveness

after oral challenge with OVA. We have found that Formula-3 significantly reduced IgE production by inhibiting Th2 cytokine release. On the other hand, Formula-3 showed potential to block mast cell degranulation by decreasing Ca^{2+} influx via SOC channel.

Mast cells are the main effector cells in the pathogenesis of multiple allergic diseases, including asthma, allergic rhinitis, gastrointestinal allergy, and cutaneous anaphylaxis. The majority of mast cell studies have addressed their predominant role in acute allergic reactions (immediate hypersensitivity) and, more recently, their roles in late-phase allergic reactions [41,45]. Therefore, developing new drugs capable of stabilizing mast cells would be valuable for treating diseases attributable to type I hypersensitivity reactions. Using RBL-2H3 cell line, intensive researches have been focused on looking for promising drugs to inhibit mast cell activation and degranulation, among which Formula-1 [19], -2 [20,33] and some of their individual recipes [33] have latent effects. For example, a Chinese herbal formula, Food Allergy Herbal Formula-2 composed of nine herbs, blocked peanut-induced anaphylaxis in a murine model [20]. In the present study, we found that modified Chinese herbal medicine, Formula-3, strongly suppressed IgE-mediated mast cell degranulation, suggesting potential therapeutic application of Formula-3 for treating allergic diseases. This finding was confirmed by using a mast cell-dependent food allergic rat model, in which Formula-3 administration inhibited OVA-sensitized mast cell activation and degranulation. Moreover, it was demonstrated in this study that application with Formula-3 at a high concentration (0.8 mg/ml, which is 4 times of that for mast cell stabilization) did not show cytotoxicity on RBL-2H3 cells. This finding may shed some light that Formula-3 is a good candidate for in vivo treatment of allergic diseases.

The importance of calcium influx in mast cell activation and degranulation has been well recognized [3]. The degranulation of mast cell is Ca^{2+} dependent and an increase in intracellular Ca^{2+} characterized by Ca^{2+} entry through SOCs is essential for granule release [11,28,35]. In this study, we found that Formula-3 treatment significantly

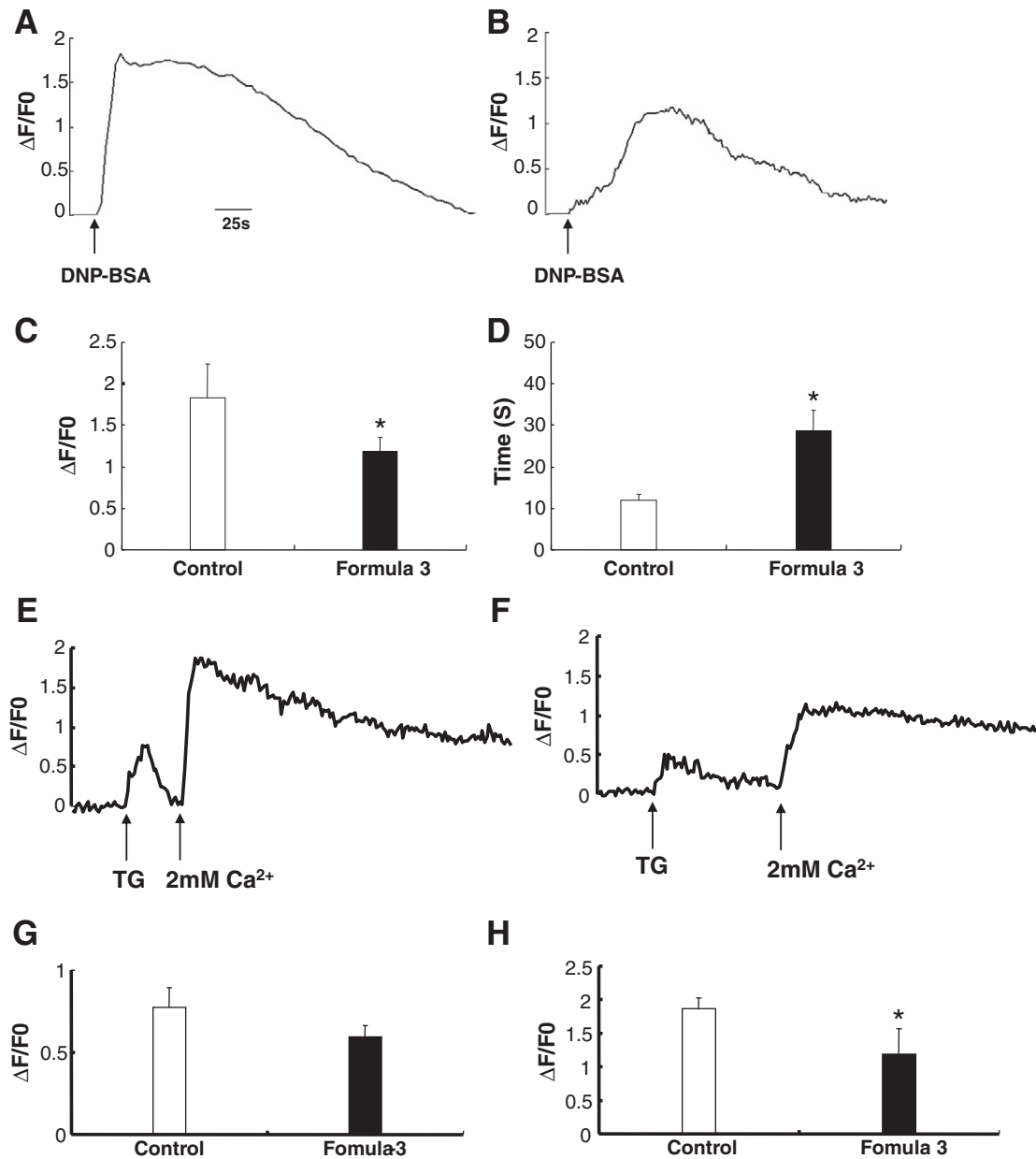


Fig. 7. Formula-3 suppressed Fc ϵ RI-mediated Ca^{2+} . A–B, A typical Ca^{2+} response to Fc ϵ RI activation in control and Formula-3-treated cells. C, Statistical analysis of the amplitude of the Fc ϵ RI-induced Ca^{2+} increase in both groups. The amplitude of the Ca^{2+} response was represented as the highest observed level of $\Delta F/F_0$. D, Statistical analysis of the time needed to reach the highest concentration of intracellular Ca^{2+} in both groups. E–F, Typical time course of the Ca^{2+} response following TG stimulation with Ca^{2+} add-back in control (E) and Formula-3-treated cells (F). The first Ca^{2+} peak was stimulated by TG under 0 mM Ca^{2+} , which represents the ER Ca^{2+} release. The second Ca^{2+} peak was elicited with 2 mM Ca^{2+} added back in the presence of TG, which represents Ca^{2+} entry through SOCs. G–H, The average amplitude of the first (G) and second (H) Ca^{2+} increases in control and Formula-3 groups. *, $P < 0.05$ vs Ctrl.

attenuated Fc ϵ RI-elicited intracellular Ca^{2+} increase, indicating that Formula-3 stabilizes mast cell by suppressing Ca^{2+} mobilization. Furthermore, we found that Formula-3 inhibited Ca^{2+} entry through SOCs. Multiple mechanisms are involved in the regulation of SOC activity: 1) stimuli that release Ca^{2+} from the ER, including receptors that activate phospholipase C (PLC) to produce inositol 1,4,5-trisphosphate (IP_3) and activate IP_3 receptors; and 2) direct regulation on the two respective limiting components, STIM1 and Orai1 [23,30]. It has recently been discovered that the two subunits, STIM1 and Orai1, play vital roles in both the signaling and the permeation mechanisms for Ca^{2+} influx through SOCs. Overexpression of STIM1 together with Orai1 caused a dramatic increase in store-operated Ca^{2+} entry in RBL cells [37]. The mechanism underlying Formula-3 inhibition of SOC activity remains unknown.

In summary, the present study established Formula-3 as a novel mast cell stabilizer, with the capacity for preventing pathogenesis of food allergy and perhaps other mast cell-dependent allergic diseases through stabilizing mast cells. The underlying mechanism for Formula-3 induced stabilization of mast cells is related to inhibition of Ca^{2+} mobilization upon Fc ϵ RI activation.

Acknowledgments

This work was supported by grants from the Natural Science Foundation of China (no. 81271950 to Q.J.), Projects of International/HMT (Hong Kong, Macau and Taiwan) Cooperation and Innovation Platform in Science and Technology of Guangdong Higher Education Institutions (no. 2012gjhz0009 to Z.L.), Key Laboratory Construction

Program of Shenzhen (no. SW201110010 to Z.L.) and Basic Research Program of Shenzhen University (no. 201101 to Z.L.).

References

- [1] Ahui ML, Champy P, Ramadan A, Pham Van L, Araujo L, Brou Andre K, et al. Ginger prevents Th2-mediated immune responses in a mouse model of airway inflammation. *Int Immunopharmacol* 2008;8:1626–32.
- [2] Alvarez DF, King JA, Townsley MI. Resistance to store depletion-induced endothelial injury in rat lung after chronic heart failure. *Am J Respir Crit Care Med* 2005;172:1153–60.
- [3] Blank U, Rivera J. The ins and outs of IgE-dependent mast-cell exocytosis. *Trends Immunol* 2004;25:266–73.
- [4] Boyce JA, Assa'ad A, Burks AW, Jones SM, Sampson HA, Wood RA, et al. Guidelines for the diagnosis and management of food allergy in the United States: report of the NIAID-sponsored expert panel. *J Allergy Clin Immunol* 2010;126:S1–S58.
- [5] Burks AW, Tang M, Sicherer S, Muraro A, Eigenmann PA, Ebisawa M, et al. ICON: food allergy. *J Allergy Clin Immunol* 2012;129:906–20.
- [6] Chan CK, Kuo ML, Shen JJ, See LC, Chang HH, Huang JL. Ding Chuan Tang, a Chinese herb decoction, could improve airway hyper-responsiveness in stabilized asthmatic children: a randomized, double-blind clinical trial. *Pediatr Allergy Immunol* 2006;17:316–22.
- [7] Chen ML, Lin BF. Effects of triterpenoid-rich extracts of *Ganoderma tsugae* on airway hyperactivity and Th2 responses in vivo. *Int Arch Allergy Immunol* 2007;143:21–30.
- [8] Chen XW, Lau KW, Yang F, Sun SS, Fung MC. An adjuvant free mouse model of oral allergenic sensitization to rice seeds protein. *BMC Gastroenterol* 2011;11:62.
- [9] Choo MK, Park EK, Han MJ, Kim DH. Antiallergic activity of ginseng and its ginsenosides. *Planta Med* 2003;69:518–22.
- [10] De Jonge F, De Laet A, Van Nassauw L, Brown JK, Miller HR, van Bogaert PP, et al. In vitro activation of murine DRG neurons by CGRP-mediated mucosal mast cell degranulation. *Am J Physiol Gastrointest Liver Physiol* 2004;287:G178–91.
- [11] Fewtrell C, Sherman E. IgE receptor-activated calcium permeability pathway in rat basophilic leukemia cells: measurement of the unidirectional influx of calcium using quin2-buffered cells. *Biochemistry* 1987;26:6995–7003.
- [12] Ghannadan M, Baghestanian M, Wimala F, Eisenmenger M, Latal D, Kargul G, et al. Phenotypic characterization of human skin mast cells by combined staining with toluidine blue and CD antibodies. *J Invest Dermatol* 1998;111:689–95.
- [13] Grundy J, Matthews S, Bateman B, Dean T, Arshad SH. Rising prevalence of allergy to peanut in children: data from 2 sequential cohorts. *J Allergy Clin Immunol* 2002;110:784–9.
- [14] Gupta R, Sheikh A, Strachan DP, Anderson HR. Time trends in allergic disorders in the UK. *Thorax* 2007;62:91–6.
- [15] Hsu CH, Lu CM, Chang TT. Efficacy and safety of modified Mai-Men-Dong-Tang for treatment of allergic asthma. *Pediatr Allergy Immunol* 2005;16:76–81.
- [16] Huang F, Yamaki K, Tong X, Fu L, Zhang R, Cai Y, et al. Inhibition of the antigen-induced activation of RBL-2H3 cells by sinomenine. *Int Immunopharmacol* 2008;8:502–7.
- [17] Itoh T, Tsukane M, Koike M, Nakamura C, Ohguchi K, Ito M, et al. Inhibitory effects of whisky congeners on IgE-mediated degranulation in rat basophilic leukemia RBL-2H3 cells and passive cutaneous anaphylaxis reaction in mice. *J Agric Food Chem* 2010;58:7149–57.
- [18] Jia XD, Li N, Wu YN, Yang XG. Studies on BN rats model to determine the potential allergenicity of proteins from genetically modified foods. *World J Gastroenterol* 2005;11:5381–4.
- [19] Kalliomaki M, Salminen S, Arvilommi H, Kero P, Koskinen P, Isolauri E. Probiotics in primary prevention of atopic disease: a randomised placebo-controlled trial. *Lancet* 2001;357:1076–9.
- [20] Kattan JD, Srivastava KD, Zou ZM, Goldfarb J, Sampson HA, Li XM. Pharmacological and immunological effects of individual herbs in the Food Allergy Herbal Formula-2 (FAHF-2) on peanut allergy. *Phytother Res* 2008;22:651–9.
- [21] Knippels LM, Penninks AH, Spanhaak S, Houben GF. Oral sensitization to food proteins: a Brown Norway rat model. *Clin Exp Allergy* 1998;28:368–75.
- [22] Lee K, Sohn Y, Lee MJ, Cho HS, Jang MH, Han NY, et al. Effects of *Angelica acutiloba* on mast cell-mediated allergic reactions in vitro and in vivo. *Immunopharmacol Immunotoxicol* 2012;34:571–7.
- [23] Lewis RS. Store-operated calcium channels: new perspectives on mechanism and function. *Cold Spring Harb Perspect Biol* 2011;3.
- [24] Li XM, Brown L. Efficacy and mechanisms of action of traditional Chinese medicines for treating asthma and allergy. *J Allergy Clin Immunol* 2009;123:297–306 [quiz 307–298].
- [25] Li XM, Huang CK, Zhang TF, Teper AA, Srivastava K, Schofield BH, et al. The Chinese herbal medicine formula MSSM-002 suppresses allergic airway hyperactivity and modulates TH1/TH2 responses in a murine model of allergic asthma. *J Allergy Clin Immunol* 2000;106:660–8.
- [26] Li XM, Zhang TF, Huang CK, Srivastava K, Teper AA, Zhang L, et al. Food Allergy Herbal Formula-1 (FAHF-1) blocks peanut-induced anaphylaxis in a murine model. *J Allergy Clin Immunol* 2001;108:639–46.
- [27] Lin RY, Anderson AS, Shah SN, Nurrizzaman F. Increasing anaphylaxis hospitalizations in the first 2 decades of life: New York State, 1990–2006. *Ann Allergy Asthma Immunol* 2008;101:387–93.
- [28] Ma HT, Peng Z, Hiragun T, Iwaki S, Gilfillan AM, Beaven MA. Canonical transient receptor potential 5 channel in conjunction with Orai1 and STIM1 allows Sr^{2+} entry, optimal influx of Ca^{2+} , and degranulation in a rat mast cell line. *J Immunol* 2008;180:2233–9.
- [29] Montalto M, Santoro L, D'Onofrio F, Curigliano V, Gallo A, Visca D, et al. Adverse reactions to food: allergies and intolerances. *Dig Dis* 2008;26:96–103.
- [30] Muik M, Schindl R, Fahrner M, Romanin C. Ca^{2+} release-activated Ca^{2+} (CRAC) current, structure, and function. *Cell Mol Life Sci* 2012;69:4163–76.
- [31] Poulos LM, Waters AM, Correll PK, Loblay RH, Marks GB. Trends in hospitalizations for anaphylaxis, angioedema, and urticaria in Australia, 1993–1994 to 2004–2005. *J Allergy Clin Immunol* 2007;120:878–84.
- [32] Poulsen LK, Hansen TK, Norgaard A, Vestergaard H, Stahl Skov P, Bindslev-Jensen C. Allergens from fish and egg. *Allergy* 2001;56(Suppl. 67):39–42.
- [33] Qu C, Srivastava K, Ko J, Zhang TF, Sampson HA, Li XM. Induction of tolerance after establishment of peanut allergy by the Food Allergy Herbal Formula-2 is associated with up-regulation of interferon-gamma. *Clin Exp Allergy* 2007;37:846–55.
- [34] Sampson HA. Epidemiology of food allergy. *Pediatr Allergy Immunol* 1996;7:42–50.
- [35] Sanchez-Miranda E, Ibarra-Sanchez A, Gonzalez-Espinosa C. Fyn kinase controls FcεpsilonRI receptor-operated calcium entry necessary for full degranulation in mast cells. *Biochem Biophys Res Commun* 2010;391:1714–20.
- [36] Slader CA, Reddel HK, Jenkins CR, Armour CL, Bosnic-Anticevich SZ. Complementary and alternative medicine use in asthma: who is using what? *Respirology* 2006;11:373–87.
- [37] Soboloff J, Spassova MA, Tang XD, Hewavitharana T, Xu W, Gill DL. Orai1 and STIM reconstitute store-operated calcium channel function. *J Biol Chem* 2006;281:20661–5.
- [38] Song Y, Qu C, Srivastava K, Yang N, Busse P, Zhao W, et al. Food Allergy Herbal Formula 2 protection against peanut anaphylactic reaction is via inhibition of mast cells and basophils. *J Allergy Clin Immunol* 2010;126(1208–1217):e1203.
- [39] Srivastava KD, Kattan JD, Zou ZM, Li JH, Zhang L, Wallenstein S, et al. The Chinese herbal medicine formula FAHF-2 completely blocks anaphylactic reactions in a murine model of peanut allergy. *J Allergy Clin Immunol* 2005;115:171–8.
- [40] Srivastava KD, Qu C, Zhang T, Goldfarb J, Sampson HA, Li XM. Food Allergy Herbal Formula-2 silences peanut-induced anaphylaxis for a prolonged posttreatment period via IFN-gamma-producing CD8+ T cells. *J Allergy Clin Immunol* 2009;123:443–51.
- [41] Swedin L, Ellis R, Neimert-Andersson T, Ryrfeldt A, Nilsson G, Inman M, et al. Prostaglandin modulation of airway inflammation and hyperresponsiveness in mice sensitized without adjuvant. *Prostaglandins Other Lipid Mediat* 2010;92:44–53.
- [42] Swindle EJ, Metcalfe DD, Coleman JW. Rodent and human mast cells produce functionally significant intracellular reactive oxygen species but not nitric oxide. *J Biol Chem* 2004;279:48751–9.
- [43] Untersmayr E, Jensen-Jarolim E. Mechanisms of type I food allergy. *Pharmacol Ther* 2006;112:787–98.
- [44] Yang C, Mo X, Lv J, Liu X, Yuan M, Dong M, et al. Lipopolysaccharide enhances Fc(εpsilon)RI-mediated mast cell degranulation by increasing Ca^{2+} entry through store-operated Ca^{2+} channels: implications for LPS exacerbating allergic asthma. *Exp Physiol* 2012.
- [45] Yuan M, Li J, Lv J, Mo X, Yang C, Chen X, et al. Polydatin (PD) inhibits IgE-mediated passive cutaneous anaphylaxis in mice by stabilizing mast cells through modulating Ca^{2+} mobilization. *Toxicol Appl Pharmacol* 2012;264:462–9.

Update

International Immunopharmacology

Volume 18, Issue 1, January 2014, Page 212

DOI: <https://doi.org/10.1016/j.intimp.2013.12.005>



Corrigendum

Corrigendum to “Herbal Formula-3 inhibits food allergy in rats by stabilizing mast cells through modulating calcium mobilization” [Int. Immunopharmacol. 17 (2013) 576–584]



Bo Yang ^{a,d,1}, Jianjie Li ^{a,d,1}, Xiaoyu Liu ^{a,d}, Li Ma ^{a,d}, Li Deng ^b, Jie Liu ^c, Zhigang Liu ^{a,d,*,1}, Qiongmei Ji ^{d,*,1}

^a State Key Laboratory of Respiratory Disease for Allergy, School of Medicine, Shen Zhen University, Shen Zhen 518060, China

^b College of Life Sciences, Shen Zhen University, Shen Zhen 518060, China

^c Department of Pathophysiology, School of Medicine, Shen Zhen University, Shen Zhen 518060, China

^d Institute of Allergy and Immunology, School of Medicine, Shen Zhen University, Shen Zhen 518060, China

The authors regret that there was a mistake in the author list. Both Zhigang Liu and Qiongmei Ji should be corresponding authors. The authors would like to apologise for any inconvenience caused.

DOI of original article: <http://dx.doi.org/10.1016/j.intimp.2013.06.016>.

* Corresponding authors at: Rm722, Composite Building, School of Medicine, Shen Zhen University, Nan Hai Ave 3688, Shen Zhen, China.
E-mail addresses: lzg195910@126.com (Z. Liu), jqm196910@126.com (Q. Ji).

¹ Both the authors have contributed equally to this work.